

Industrial Motor Troubleshooting Guide

Synthetic project reference document for Phase 5 RAG testing.

1. Motor Monitoring

Motor current, temperature, RPM, vibration, and supply conditions can provide useful indicators of motor or driven-equipment problems. Motor current should be interpreted in relation to load and machine operating conditions.

2. High Motor Temperature

Potential causes include overload, poor ventilation, cooling-system problems, excessive friction, bearing problems, electrical faults, or operation outside the rated duty. Check load conditions, cooling paths, bearing condition, current trend, and maintenance records.

3. High Motor Current

Potential causes include mechanical overload, increased friction, shaft misalignment, bearing degradation, incorrect electrical conditions, or process changes. A simultaneous rise in current and vibration may indicate increasing mechanical resistance.

4. Low or Unstable RPM

Possible causes include overload, drive-system problems, controller faults, mechanical resistance, coupling issues, or abnormal process load. Compare RPM with motor current and vibration.

5. Troubleshooting Sequence

First verify safe operating conditions. Review sensor trends, inspect cooling and ventilation, check mechanical load, inspect bearings and coupling, and review electrical diagnostics according to the approved procedure.

Important: This is synthetic reference content created for software testing and demonstration. It is not a manufacturer manual or a substitute for approved maintenance procedures, qualified personnel, or applicable safety requirements.